Ninety-Three Per Cent of a Backlog Is Not a Backlog: Reading a Public Publisher's Own Quality Signal

Abstract

Compliance systems and models grounded in statutory text now read published law directly. A companion study proposed a five-field provenance schema so that a consumer of such data can tell what stands behind each record; this article applies it to a second jurisdiction. The United Kingdom's official legislation service publishes, per Act and machine-readably, the amendments it has not applied to the text it displays: 14,407 across the Public General Acts of 1988–2026. Read as a backlog — the reading the flag invites, and the one this study first recorded — that overstates by a factor of 131. 93.0 per cent are enacted but not commenced; a further 901 carry a commencement date still in the future, and the service is right to apply neither. What remains is 99 effects in 9 Acts, none in force longer than 21 days against a published three-month target, all from one commencement three weeks before the analysis date. The service meets its own standard without exception. The transferable result is the error: everything needed to read the record correctly is published, but the flag inviting the wrong conclusion is an attribute on each effect while the fact correcting it is a child element inside it, and the publisher's prose covers all four states with one term. A consumer who parses attributes and checks the documentation is confirmed in a false accusation against a service doing its job. Publishing the distinguishing fact is not enough; its placement decides what most readers conclude.

Keywords: public data infrastructure; data quality; open government data; metadata; legal informatics; provenance

1 Introduction

A companion study measured whether one state's official legal record was available and internally consistent when someone tried to build on it, and found that neither could be assumed.[1] It proposed a five-field provenance schema for publishers of official data, and stated a weakness plainly: one jurisdiction, and an argument rather than a demonstration that the method travels.

This article is the demonstration, and what it demonstrates is not what it set out to. Repeating a measurement elsewhere tests whether a number recurs. Applying an instrument elsewhere tests the instrument. The instrument failed first, in a way worth reporting.

The United Kingdom's official legislation service publishes, in each Act's own metadata, a machine-readable list of amendments it has not applied to the text it displays. Read as a backlog, that list says 14,407 amendments across the modern statute book are missing from the law as published — an unusually clean finding, and the one this study first recorded.

It is wrong by a factor of 131, and it took two corrections rather than one to find out — which is itself part of the argument. Each effect also carries whether the amendment is in force. 13,396 of them (93.0 per cent) are prospective: enacted, not commenced, no date set. A further 901 have a commencement date still in the future. The service is right not to apply either — applying an amendment that is not in force would misstate the law. What remains is 99 effects, and none has been in force for more than 21 days against a published target of three months.

[1] Cited as a companion study so this article can be read on its own; the corpus and analysis are openly published.

The second of those categories is why one correction was not enough. Excluding the prospective effects is the obvious fix once the in-force field is found, and it still overstates by an order of magnitude: an effect can be non-prospective — its commencement settled — with that commencement fixed for a date that has not arrived. A date set for next spring is no more in force than no date at all. Each stage of this measurement looked correct until the next field was read.

The finding, then, is the opposite of the one this article was drafted to report: the service meets its own standard without exception. But the more useful result is the error. The distinguishing field is a child element inside each effect; the flag that invites the wrong reading is an attribute on the effect itself, and the publisher's own prose calls both 'unapplied effects' without separating them. A reader who takes the obvious path — parse the attributes, trust the wording — produces a false accusation against a public service that is doing its job, and produces it with every appearance of rigour.

That is the companion study's argument arriving from an unexpected direction. That paper asked publishers to ship the strength of the evidence with the text. Here the evidence is shipped, completely and machine-readably, and the default reading of it is still wrong by two orders of magnitude. Publishing the distinguishing fact is not sufficient; where it sits, relative to the fact that invites the error, decides what most readers will conclude.

2 Related work

Three literatures bear on this article, and it sits in a gap between them.

Data quality as fitness for use. The framing that treats quality as a property of use rather than of the artifact goes back to Wang and Strong (1996), whose survey of data consumers found a conception of quality much broader than accuracy, organized into dimensions including believability, interpretability and ease of understanding. That framework anticipates the problem here in one respect and misses it in another. It is right that quality is relative to a consumer; but its dimensions are properties a dataset *has*, assessed one at a time. The record examined in this article scores well on every one of them and still produces a wrong answer by default, because the fault is in the relation between two fields rather than in either field.

Open government data and metadata quality. The open-data literature has repeatedly identified metadata as the binding constraint on reuse. Janssen et al. (2012) cataloged adoption barriers and argued against a 'publish and they will come' model of open data. Neumaier et al. (2016) built the closest methodological antecedent to this study: an automated, repeated assessment of metadata quality across many open data portals, measuring completeness, accuracy and retrievability at scale.

That work measures whether metadata is present and correct. This article assumes both — the metadata examined here is present, correct, complete and machine-readable — and asks a different question: whether a competent consumer, reading it in the ordinary way, arrives at what it says. The two questions come apart, and the second has no established measure.

Provenance and dataset documentation. The W3C PROV data model gives a general vocabulary for recording what an artifact was derived from and by what activity (W3C, 2013). Gebru et al. (2021) proposed a documentation practice — a prose record of motivation, composition, collection process and recommended uses — to travel with a dataset and make its limits legible to a later user. Both address the case where the needed information is *absent*. Neither addresses the case examined here, where the information is present, structured and authoritative, and the consumer still gets it wrong because of where it sits.

The gap. Between 'the publisher did not say' and 'the consumer did not read' there is a third condition: the publisher said it, in a place and a vocabulary that make the wrong reading the cheap one. This article is an account of that condition, measured on a national statute book, together with four properties by which a schema can be checked for it.

3 What the service publishes about its own currency

Two features of `legislation.gov.uk` matter here, and both are the publisher's own statements rather than this study's inferences.

The first is the *unapplied effect*. When an amendment has been made to an Act and the revised text has not yet been changed to reflect it, the service records the fact in the Act's metadata and flags it to readers: 'If the primary legislation on legislation.gov.uk has any unapplied effects, we flag them in the "Changes to Legislation" banner on the website. Clicking on the banner reveals the outstanding changes.'[2] Each such effect names the amending instrument, the provision affected, and the kind of change.

The second is a service standard: 'We aim to incorporate new amendments into the text of the legislation within three months of those amendments coming into force.'[3]

The standard is what makes compliance measurable rather than a matter of impression, and it also rules something out. A three-month target means the expected steady state is not zero unapplied effects: a queue is the design, not the defect, so the existence of flagged effects establishes nothing on its own. What the standard permits is a direct test — how long has each amendment actually been in force without being applied — and the answer needs the commencement date, which the service publishes for every effect.

4 Method

Collection. Every Public General Act from 1988 to 2026 was enumerated from the service's own year feeds, and each Act's metadata retrieved from its `introduction/data.xml` endpoint. That endpoint carries the complete unapplied-effects block — verified rather than assumed, by comparing it against the full text and the contents listing for the same Act and finding the same effects in all three — at a fraction of the bytes. The service's `robots.txt` declares a five-second crawl delay and the collector honours it, which is why a sweep of the statute book takes hours.

The schema, populated at collection time. The companion study's five fields were written for each record as it was retrieved, unchanged from that paper's definition. This is the point of the exercise rather than decoration: that paper claims the schema costs nothing because a collector that knows how it obtained a document already knows all five values. Populating them prospectively is the only way to find out whether that is true.

What is measured, and what is not. The companion study measured two properties: availability and internal consistency. Only the second is compared here, and the reason is structural rather than practical. Four of the schema's five fields describe the *encounter* — whether the live source answered this collector, on this network, on this day — and encounters do not compare across collections, because they differ before the records do. No control corrects for that. Availability figures are recorded and published for this collection, and are not set against the companion study's.

An unapplied effect has no such problem *in principle*: it is a property of the data as published, so any reader, on any network, at any time, can obtain the same count. In practice they will not, and the rest of this article is about why — the count depends on which of the record's fields the reader consults, and the obvious selection is wrong by two orders of magnitude. Reproducibility of a measure and agreement between measurers are different properties, and only the first is guaranteed here.

[2] legislation.gov.uk (2026), under the heading 'Our editorial practice and timescales'. The heading matters: the same page carries a separate section on legislation originating from the EU, and the commitment quoted here sits outside it and is stated generally.

[3] legislation.gov.uk (2026), same heading.

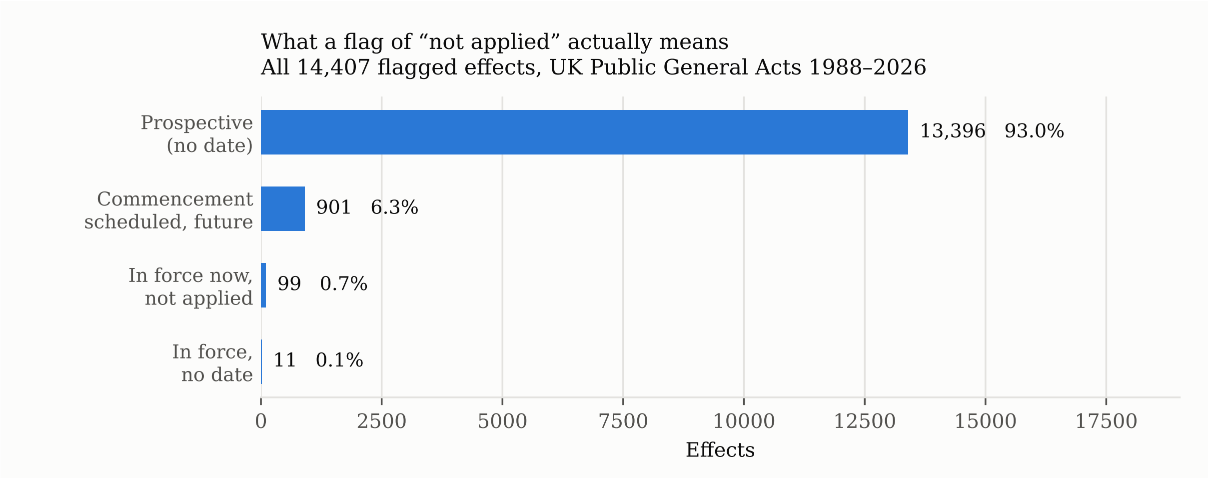

Figure 1: The four states of a flagged effect, on a linear scale. The scale is linear because the disproportion is the finding: a logarithmic axis would fit all four bars comfortably and remove it.

5 Results

Coverage. 1,578 Public General Acts from 1988 to 2026 were listed and 1,578 retrieved; none was lost.

What the flag contains. Of the 14,407 effects the service flags as not applied:

	Effects	Share
Prospective — enacted, no commencement date	13,396	93.0%
Commencement scheduled, date still future	901	6.3%
In force now, not applied	99	0.7%
In force, no date recorded	11	—

Table 1: The same flag, four meanings. Only the last two describe text that does not reflect the law.

The measure that matters. 9 Acts of 1,578 — 0.6 per cent — display text that does not reflect the law in force, across 77 distinct provisions. Against a body of 101,711 paragraphs, that is 0.8 affected provisions per thousand.

Compliance is total, and concentrated in one event. The service undertakes to incorporate amendments within three months of their coming into force. 0 effects exceed that. The longest any amendment in the modern statute book has been in force without being applied is 21 days.

The dated in-force effects are not spread across the period at all. All 99 came into force on the same day, three weeks before the analysis date, and all belong to two related instruments: the Pension Schemes Act 2021 and the Pension Schemes (Northern Ireland) Act 1993.[4] The entire population of out-of-date text in the modern UK statute book is one commencement, three weeks old.

That is why this article carries one figure and not two. A distribution of 99 identical values is not a chart.

[4] The remaining 11 effects are recorded as in force without a date and cannot be timed; they are counted in the affected totals and excluded from the duration figures.

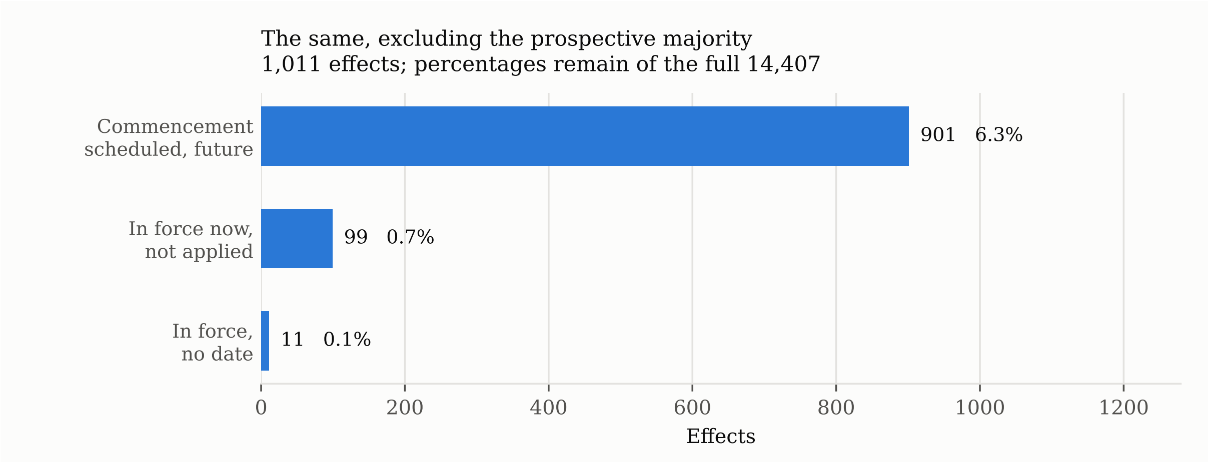

Figure 2: The same effects excluding the prospective majority, where the three remaining categories become legible. Percentages are still of the full population, so the two figures can be read against each other.

6 Availability, recorded and not compared

The companion study's headline measures were about availability: whether the official source answered at all. This collection recorded the same measures and does not set them against that study's, for a reason that is a finding about the schema rather than a limitation of the data.

Of 1,578 Acts listed, 1,578 were retrieved. Retrieval failures during collection were recorded rather than retried away while availability was still in scope; a second pass over them recovered every one, each in under a second, and both attempts are kept in the record. Successful responses arrived in under two seconds throughout. Every failure was a connection that produced no bytes and then expired.

Those figures describe this collector, on this network, over this period. They do not describe `legislation.gov.uk`, and they cannot be compared with a collection made from a different network at a different time. The companion study's Saudi measurements were taken from the author's own connection, over an earlier period, against portals whose behavior has since changed — and that measurement cannot be repeated, which is precisely the argument that study makes about contemporaneous evidence.

A control for the network would address half of that and leave the period untouched. The comparison is therefore not attempted. What is reported instead is the structural reason it cannot be, developed in section 10: four of the schema's five fields describe the encounter rather than the record, and encounters differ before records do.

7 Reproducibility

Every number in this article is produced by scripts published with it, and the article contains no typed figure: the manuscript source carries macros, and the macros are regenerated from the analysis results before each compilation. A figure in the prose therefore cannot drift from the figure in the data, because only one copy of it exists. A separate check refuses to compile a manuscript that types a number the analysis owns.

The analysis is verified against a second implementation written to share no code with it, recounting the headline figures from the raw collection by the most direct route available. That check is reported here with a caveat earned during this study: it agreed with the analysis at every stage, including the two that were wrong. Two implementations of the same definition agree by construction. Such a check detects arithmetic and bookkeeping faults and cannot detect a mistaken definition, which is the class of fault that actually threatened this paper.

The collection is reproducible in the ordinary sense — the collector is published, the endpoints are stable, and a year already gathered is skipped so a sweep can resume — but it is a snapshot. The service revises its records continuously; `dc:modified` on the affected Acts shows revisions within weeks of the analysis date. A collection taken later will differ, and the figures here are stamped with the date they describe.

8 Three measurements of the same thing

The measure reported above was reached in three stages, and the two discarded stages are more instructive than the surviving one. Each was defensible on the evidence available when it was made, each produced a number that survived internal checking, and each was wrong.

First: the flag alone. `RequiresApplied="true"` is an attribute on each effect. Read as 'this amendment should be in the text and is not', it gives 14,407 effects. That reading was tested in the two ways available without further parsing. It was checked against the publisher's own description, which speaks of flagging 'unapplied effects' and does not distinguish states beneath that phrase. And it was checked against an independent recount, written to share no code with the analysis, which agreed on every figure.

Neither check could have caught the error. The documentation's wording is true of all four states, so consulting it confirms any of them. The independent recount counted the same wrong thing by a different route, which is what an independent implementation of a wrong definition does. Agreement between two implementations tests the implementations, not the definition.

Second: excluding the prospective effects. `ukm:InForce` is a child element inside each effect, invisible to an attribute-level parse. Reading it and excluding `Prospective="true"` removes 13,396 effects and leaves 1,011. This is the obvious correction, and it is the one most consumers who discover the field would make.

It still overstates by an order of magnitude. An effect can be non-prospective — its commencement settled — with the commencement date itself still in the future. 901 effects are in exactly that position. The field that distinguishes them is not a different element but a different attribute of the same one, and its meaning depends on a comparison with the date of reading.

Third: comparing the commencement date with the present. This is the measure reported above: 110 effects in 9 Acts.

What the sequence shows. Each stage required reading one level deeper into the same response, and at each stage the previous answer looked complete. There was no error message, no missing field, no malformed record — the data was correct and completely published throughout. What changed was how much of it was consulted.

This is the difference between a record being *available* and a record being *read correctly*, and it is not addressed by any of the measures usually proposed for open data quality. Completeness, timeliness, accuracy and machine-readability were all satisfied at every stage. A dataset can score perfectly on each and still have a default reading that is wrong by two orders of magnitude.

9 What made this record misreadable

The failure here is structural rather than accidental, and the structure can be described in a way that transfers to other datasets.

The inviting fact is cheaper to reach than the correcting fact. `RequiresApplied` is an attribute on the effect element. `ukm:InForce` is a child element of it. In every common way of consuming XML — an attribute map, a flattening to tabular form, a regular expression over tags — the first is obtained by default and the second requires a deliberate second step. Cost asymmetry between a fact and its qualifier determines which one a consumer ends up with.

The qualifier is not marked as required. Nothing in the record indicates that `RequiresApplied` is uninterpretable on its own. A consumer who reads it and stops receives no signal that anything is missing, because from the parser's point of view nothing is.

The vocabulary is coarser than the schema. The publisher's documentation calls all four states 'unapplied effects'. The term is accurate for each of them and discriminating for none. Prose that names a superset of what the data distinguishes will validate a reader's wrong reading, and the more carefully that reader checks the documentation, the more confident the wrong reading becomes.

The consequence is asymmetric. Reading `RequiresApplied` alone does not produce a slightly imprecise figure. It produces one 131 times too large, pointing in the direction of accusation, against an institution that is meeting its published commitment. The error is not merely large; it is large in the direction that generates findings people want to publish.

None of these four properties is unique to this dataset. Together they describe a shape — an inviting attribute, a qualifying child, an undiscriminating label — that a publisher can look for in its own schema, and that a consumer can look for before trusting a count.

10 What the schema learned about itself

Two changes to the companion study's proposal came out of using it, and neither would have come out of thinking about it.

A discrepancy has to record who declared it. Applied to a well-run interface, four of the five fields take a single value across every record in the collection: one source class, one retrieval route, one corroboration count, one transformation. That is the schema working, not failing — it costs nothing here because there is nothing to say.

But the defect this study measures was invisible to all five fields. Every affected Act was retrieved perfectly — official primary source, live, no transformation, nothing to report — so under the schema as written an Act displaying text that does not reflect the law is indistinguishable from one that is entirely current. The argument does not depend on the size of the defect, and the size is small: 9 Acts, 110 effects. An earlier draft made the same point with an Act omitting hundreds of amendments, which the corrected measure shows does not exist.

The remedy is not a sixth field. The purpose the companion study gives `discrepancy` is 'is there a known problem with this record?', and a publisher flagging its own text as not current is exactly that. What the field's *values* omit is that a discrepancy can be declared by the source as well as found by the collector — and a null means different things in the two cases. Found by a collector, it cost two or more sources and a comparison, so a null may only mean nobody looked hard enough. Declared by a source, it costs nothing to record, so a null means the publisher did not say. A single null standing for both is the collapse that paper objects to when one confidence score stands for availability and consistency at once.

The two jurisdictions are the argument. Here, 9 records carry a source-declared discrepancy and 0 a collector-found one. In the companion corpus it was the reverse: four self-contradicting instruments, every one of them found by hand. The same class of defect, with opposite provenance — and a straightforward consequence for who should be doing the work.

The fields divide by what they describe, not by what they are about. `source_class` describes the record: whose copy this is. `retrieval_route`, `corroboration` and `transformation` describe the encounter: whether the live source answered *us*, how many sources *we* found agreeing, what *we* had to do to the bytes we got. `discrepancy` is either, depending on who declared it.

Four of five describe the encounter, which is why the availability half of this comparison could not be salvaged. It also complicates the companion study's own recommendation. That paper asks publishers to populate these fields; if a publisher does, the encounter fields describe the publisher's encounter with its own source, which is a third thing again and comparable with neither. A provenance schema intended for use across collections has to mark which of its fields are encounter-relative. That one does not.

11 What follows

Publishing the distinguishing field is not enough; placing it is. Everything needed to read this record correctly is published. The failure is one of arrangement: the flag that invites the wrong conclusion is an attribute on the effect, and the fact that corrects it is a child element inside it. A consumer parsing attributes — the ordinary way to read this format — gets the first and not the second, and nothing in the response marks the omission. A publisher cannot control how it is read, but it can decide which fact is cheapest to reach.

Prose that does not distinguish what the data distinguishes. The service describes flagging 'unapplied effects'. Its data separates four states under that phrase, only two of which mean the text is out of date. The documentation was checked against the reading used here and appeared to confirm it, because the words are true of all four. Where a vocabulary is coarser than the schema it describes, the documentation will validate a wrong reading.

A measurement that reverses is worth more than one that confirms. This study set out expecting to find a backlog, found one, and had to withdraw it. The corrected result — a public publisher meeting its stated standard without exception across 1988–2026 — is a kind of finding the open-data literature reports rarely, in part because it is not what anyone goes looking for.

12 Implications

The diagnosis in section 5 is of no use unless it tells someone what to do. It gives different instructions to the two parties, and the instruction to the publisher is the cheaper one.

12.1 For publishers of official data

Make the qualifier as cheap to reach as the fact it qualifies. The concrete change here is small: promote the in-force state to an attribute on the effect element, alongside the flag it governs, or add an attribute that summarizes it. Nothing new would be published; the same fact would simply cost the same to obtain as the one that misleads without it. Where a schema cannot be changed, the same effect is available by publishing a derived field — a single attribute whose value is the state a correct reading produces.

Name the states in the documentation. The phrase 'unapplied effects' is accurate for four distinct conditions and discriminating for none. A sentence naming them, and saying which mean the displayed text is out of date, would have prevented the error reported here at the cost of one sentence. A vocabulary coarser than the schema does not merely fail to help; it actively confirms a wrong reading for the reader who checks.

Publish the count you would want cited. A publisher that knows how its record is likely to be misread can pre-empt the misreading by publishing the aggregate itself. Had the service published 'amendments in force and not yet applied: 99', no consumer would have needed to derive it, and no consumer could have derived it wrongly. This is not a matter of trusting the publisher's arithmetic over one's own; it is that a published aggregate is a statement the publisher can be held to, and a derived one is not.

12.2 For consumers of official data

A flag is not a measurement. The failure here began by treating a boolean as an answer. Before a flag is counted, the question is what it is *not* saying: under what conditions is it set, and what else in the record governs its meaning. That question is answerable from the record itself and takes minutes.

Independent reimplementation does not validate a definition. The verifier built for this study agreed with the analysis at every stage, including both wrong ones, because it implemented the same definition by a different route. Reimplementation catches arithmetic and bookkeeping; it cannot catch a mistaken reading, and treating agreement between two implementations as confirmation is a specific and common error.

Prefer the finding that is harder to publish. Each wrong reading here produced a result that was more publishable than the correct one: a large backlog at a well-run public service, concentrated in identifiable instruments, with an apparent trend. The correct result is that nothing is wrong. A measure that yields an unusually clean adverse finding about an institution deserves more scrutiny than one that does not, in proportion to how much its author would like it to be true.

Where an accusation is implied, verify against the accused's own account. The service publishes both a target and the data needed to test compliance with it. Checking the specific claim — has any amendment exceeded three months — would have exposed every version of the error immediately. It was not checked earlier because the aggregate looked decisive on its own.

13 Threats to validity

One jurisdiction, one publisher, one format. The structural diagnosis in section 5 is drawn from a single schema. Its four properties are stated so that they can be looked for elsewhere, not because they have been observed elsewhere.

Primary legislation only. Statutory instruments are not covered. They are the larger population, they carry their own unapplied effects, and nothing here should be read as a statement about them.

A snapshot of a moving record. All figures describe the collection date. The in-force effects reported here arose from a commencement three weeks earlier and will be applied, on the publisher's stated timetable, within another two months. A collection taken at a different moment would find different effects — and, if the service continues to meet its target, a similarly small number of them.

The undated in-force effects. 11 effects are recorded as in force with no date. They are counted in the affected totals and excluded from every duration figure, because their duration is unknown. If any of them has been in force for longer than the target, this article would not detect it. That is the one direction in which the compliance finding could be wrong, and it is bounded: 11 effects out of 14,407.

The measure could be wrong again. Three readings of this record have been reported, two of them withdrawn. Each was checked and each survived its checks. The third is reported with more confidence than the first two, on the grounds that it consults every field the record carries about the state of an effect — but that was the belief at each earlier stage as well. A reader with a fourth reading is invited to test it against the published collection.

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Declarations

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Declaration of generative AI use. *Tool:* Claude Opus 5 (Anthropic). *How it was used:* to write the collection, analysis, verification and figure scripts that produce every number reported here, to draft and revise the manuscript, and to run adversarial checks against its own claims. *Reason for use:* the study required retrieving and parsing the metadata of the entire modern statute book under a five-second crawl delay and auditing the result, which is not feasible by hand at that scale for a single researcher. The author set the research question, directed the analysis, verified the outputs against the underlying records, and is responsible for the content.

Data availability. The collection, the analysis and the scripts that reproduce every number and figure are openly available in a public repository under a permissive licence. Its address names the author, so it is given on the title page rather than here. The underlying metadata belongs to legislation.gov.uk and is published by the Crown under the Open Government Licence.

A note on the subject. This article measures a defect that legislation.gov.uk discloses about itself. It is not an accusation. Almost no comparable publisher discloses as much, which is what makes the measurement possible at all, and the article’s argument is that disclosure of this quality still leaves a reader unable to answer the question it raises.